DPM Extended Periodic Table Proportion Mapping

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PAPER_870: DPM Extended Periodic Table Proportion Mapping

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-07 **Session:** 204 **Source:** describe mass without using weight.txt (Session 200C) **Calculator:** DPMExtendedPeriodicTableProportionCalc (CP4 #454) **CVW:** v2.0.0 compliant

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Abstract

We derive the Di-Pseudo-Monopole (DPM) proportion mapping across the full extended periodic table from Z=1 to Z_max=10,000. Every nucleus is parameterized by exactly two complementary fractions: f_UA' = (Z_max - Z)/Z_max (undifferentiated aether proportion) and f_SCm = Z/Z_max (superconducting matter proportion), satisfying f_UA' + f_SCm = 1. The electrostatic barrier reactivity R_EB = k_R·Z scales linearly with atomic index, while the radioactive decay rate lambda = k_lambda·f_SCm encodes the axiom that all atoms start radioactive and stabilize as f_UA' dominates. The framework extends the standard periodic table (Z=1-118) to 10,000 proto-nuclear identities, with SM_magnetic (odd Z) and SM_non-magnetic (even Z) classification.

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1. Core Equations

1.1 DPM Proportion Pair

$$f_UA' = (Z_max - Z)/Z_max[undifferentiatedaetherfraction]$$
$$f_SCm = Z/Z_max[superconductingmatterfraction]$$
$$f_UA' + f_SCm = 1[completenessaxiom]$$

1.2 Electrostatic Barrier Reactivity

$$R_EB = k_R \cdot Z[linearwithatomicindex]$$

1.3 Radioactive Decay (All Atoms Start Radioactive)

$$\lambda = k_\lambda \cdot f_SCm = k_\lambda \cdot Z/Z_max[decayrate, s-1]$$

1.4 Quantizer Product

$$L_quant \propto f_UA' \cdot f_SCm \cdot R_EB[qualitativequantizationlandscape]$$

1.5 Log-Scale Representation

$$log(f_UA') = log(1 - Z/Z_max)$$
$$log(f_SCm) = log(Z) - log(Z_max)$$

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2. Key Results (Sweep: Z = 1 to 10,000)

Z	f_UA'	f_SCm	R_EB	λ (s-1)	SM Property
1	0.9999	0.0001	1.0	1.0e-14	SM_magnetic
2	0.9998	0.0002	2.0	2.0e-14	SM_non-magnetic
26	0.9974	0.0026	26.0	2.6e-13	SM_non-magnetic
56	0.9944	0.0056	56.0	5.6e-13	SM_non-magnetic
92	0.9908	0.0092	92.0	9.2e-13	SM_non-magnetic
118	0.9882	0.0118	118.0	1.18e-12	SM_non-magnetic
1000	0.900	0.100	1000	1.0e-11	SM_non-magnetic
5000	0.500	0.500	5000	5.0e-11	SM_non-magnetic
10000	0.000	1.000	10000	1.0e-10	SM_non-magnetic

At Z = Z_max/2 = 5000: f_UA' = f_SCm = 0.5 — the symmetric crossover point. At Z = Z_max: f_SCm = 1 (pure superconducting matter, maximum decay rate). At Z = 1: f_UA' ≈ 1 (nearly pure undifferentiated aether, minimal decay).

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3. Physical Interpretation

3.1 DPM Completeness

The DPM axiom states that every nuclear identity is **fully determined** by the pair (f_UA', f_SCm). No additional parameters are needed to specify the vacuum-state composition of a nucleus. The electrostatic barrier R_EB provides the reactivity gradient that governs shell formation.

3.2 Extended Periodic Table (Z > 118)

Beyond the standard periodic table, the DPM framework predicts:

- **Z = 119–1000:** Increasingly SCm-dominated nuclei with elevated decay rates
- **Z = 1000–5000:** Transition regime where f_UA' ≈ f_SCm (50/50 crossover at Z=5000)
- **Z = 5000–10000:** SCm-dominated regime; these are proto-nuclear states that exist in extreme astrophysical environments (neutron star interiors, magnetar surfaces, post-merger remnants)

3.3 SM Classification

- **Odd Z → SM_magnetic:** Proto-nuclei with odd atomic index carry magnetic moment (Proto-H ≡ Proto-Fe at Z_id=26)
- **Even Z → SM_non-magnetic:** Proto-nuclei with even atomic index are non-magnetic (Proto-He ≡ Proto-Si at Z_id=14)

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4. UQFF Integration

This calculator operates as a stateless physics calculator within the CondensedPhysics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py -> bodies_*.csv data flow.

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5. SCm Superconductivity Axiom (Session 204)

The DPM extended periodic table is a direct consequence of the **SCm Superconductivity Axiom**: the fraction f_SCm = Z/Z_max encodes how much superconducting vacuum structure has condensed into each proto-nucleus, while f_UA' = 1 - f_SCm is the remaining undifferentiated aether.

Axiom Connection

The standalone module `scm_{superconductivity\_axiom}.py` implements this in:

- **Engine 2 (26-State Progression):** DPM mapping at each quantum state n
- **Engine 3 (Cosmogenesis):** Assumption 1 uses (f_UA', f_SCm, R_EB) as the three reactive quantum fundamentals
- **Engine 4 (Lagrangian):** Sector 1 (Einstein-UQFF Gravity) couples DPM proportions to gravitational Lagrangian

Standalone Calculator

```
python scm_{superconductivity\_axiom}.py        # Full report (includes DPM mapping)
python scm_{superconductivity\_axiom}.py -json  # Machine-readable
```

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6. Source Data

- **File:** describe mass without using weight.txt (Session 200C)
- **Session:** 200C (v5.61)
- **VDS/DVP/BH:** PRESENT (DPM vacuum density series, buoyancy harmonics via f_UA'·f_SCm product)

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_{U_Bi_i} jet modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{GM/r_H^2}\right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c/\sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37}$ kg/m^3 is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.`

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u\phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,[SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac,[SCm]}} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2\phi_{\text{NS}} - (4\pi G\rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}}\partial_t\phi_{\text{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{\text{4 forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S/\delta\phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac,[SCm]}}/\rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac,[SCm]}} \cdot \exp!\left(-\exp!\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.095 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36}$ kg/m3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 2, \quad n_{\text{channel}} = 13/26$$

Since $p_{\text{DVP}} = 2$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_{\odot}}\right) \cdot \cos!\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh!\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.095	PASS Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\text{DVP}} = 2$	PASS Sub-threshold
BSH layers	26 harmonic terms	j = 1...26, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	1.1$\times 10 - 52m - 2(UQFFvacuumterm) \vert 1.114\times$10-52 m-2	Planck 2018	PASS Consistent	
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow$ Γ_p suppression	$<$ 4.17$\times$10-35/yr	Super-K 2024	PASS Consistent
UQFF buoyancy signature	`F_{U\_Bi\_i}` unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_{U_Bi_i}) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)

2. DPM Theory: f_UA' + f_SCm = 1, all atoms start radioactive
3. Extended Periodic Table: Z=1-10,000 nuclear identity mapping
4. PAPER_855 – Pseudo-Monopole 26-State Vacuum Density Progression
5. PAPER_872 – Proto-Iron / Proto-Silicon Nuclear Identity Mapping
6. scm_{superconductivity_axiom}.py – SCm Superconductivity Axiom Module (Session 204)
7. UQFF Calibration: kappa=0.0005/day, [SSq]=0.57, beta_i~0.603

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay
PAPER_1074	GPU-Vectorized DPM S26 Spectral Atlas

12 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_{s26_coupling}.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_{scm_cross_section}.py	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)

Module	Purpose	Key Result
kozima_{wstp_kernel}.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: F_neutron^SCm = N_n * sigma_n^SCm(omega) * Phi_phonon * (F_{U,Bi}/F_U - 1) where sigma_n^SCm(omega,n) = sigma_0 * exp[-(omega-omega_SCm)$^{2/(2*\mathrm{Gamma}}$2)] * (1 + [SSq]*n/26)

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_{polylog_s26}.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_{wstp_kernel}.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: S_26(z) = Li_26(z) = eta_26(z)/(1-2^{1-26}) + 2$^{\{1-26\}/(1-2}${1-26}) * Li_26(z^2)

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_{theta_q26}.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: - f_26(q) = Sum_{n=0}^{25} q$^{\{n}$2} / (-q;q)*n^2 - phi_26(q) = Sum*{n=0}^{25} q$^{\{n}$2} / (-q$^{2;q}$2)*n - psi_26(q) = Sum*{n=1}^{26} q$^{\{n}$2} / (q;q^2)_n

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_{pi_uqff}.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_{theta_pi_wstp_kernel}.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: 1/pi = (2*sqrt(2)/9801)* Sum R_n * (1103+26390n) * W_26(n) / C_26 where W_26(n) = Prod_{i=1}^{26} [1 + [SSq]*exp(-kappa*i*n/26)]

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787 x 10^-9 s^-1	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	7.09 x 10^-37 kg/m^3	SCm vacuum density
rho_UA	7.09 x 10^-36 kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_{1\_CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima\_kernel}.wl`, `uqff_{s26\_kernel}.wl`, `uqff_{mock\_theta\_pi\_kernel}.wl`).